

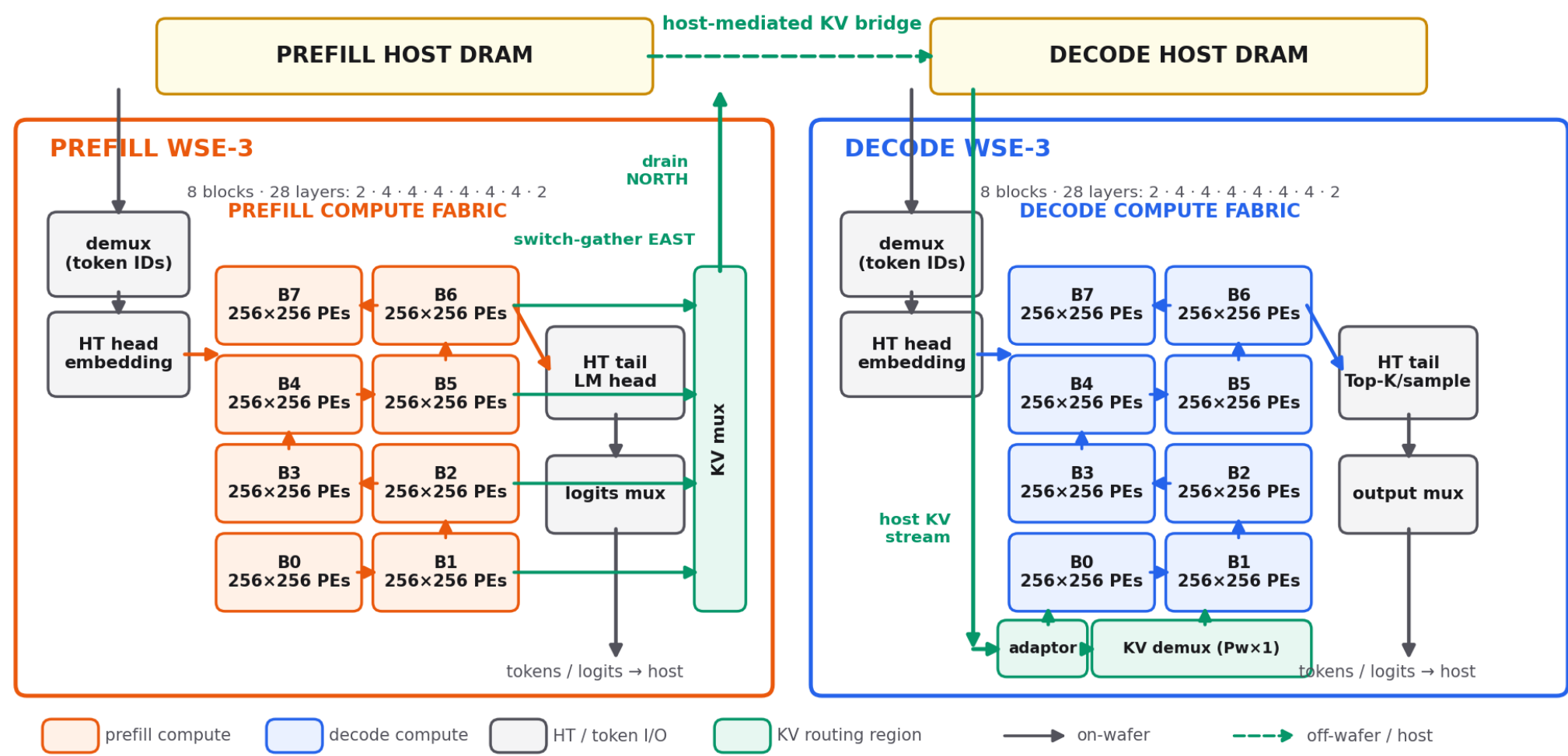
M2 STUDY · WEEKLY MEETING

KV Resume on WSE-3

Measured costs of recompute, reload, and resident-prefix prefill.

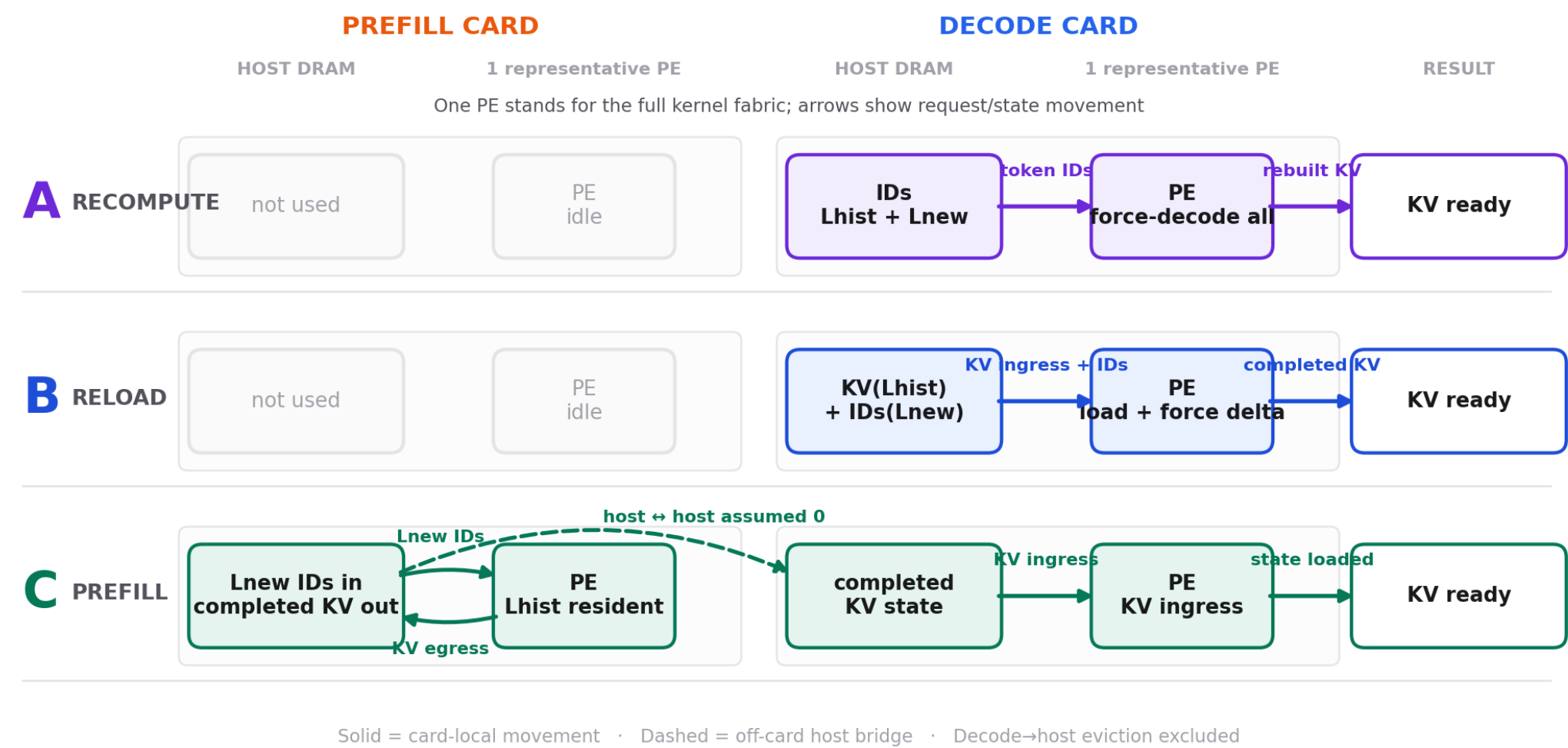
MODEL	DEVICE	TSC	SCOPE
Qwen3-1.7B	WSE-3	0.85 GHz	Resume latency

Experiment topology: prefill card → hosts → decode card



Each kernel uses a 2x4 serpentine pipeline of 256x256-PE blocks. Prefill gathers KV east into kv_mux and drains north; decode accepts the host KV stream through adaptor + kv_demux and shifts column slices north into the fabric.

Three request paths on the same simplified topology



A sends token IDs to decode and rebuilds all KV; B reloads persisted history KV and computes only Lnew; C extends resident KV on prefill, then transfers the completed state through both hosts.

Three ways to resume an evicted request

History, delta, generation

PD separation: prefill and decode each occupy one WSE card with separate host DRAM. Lhist KV is assumed persisted; Lnew has no KV. We time resume only—decode → host eviction and host-to-host latency are excluded.

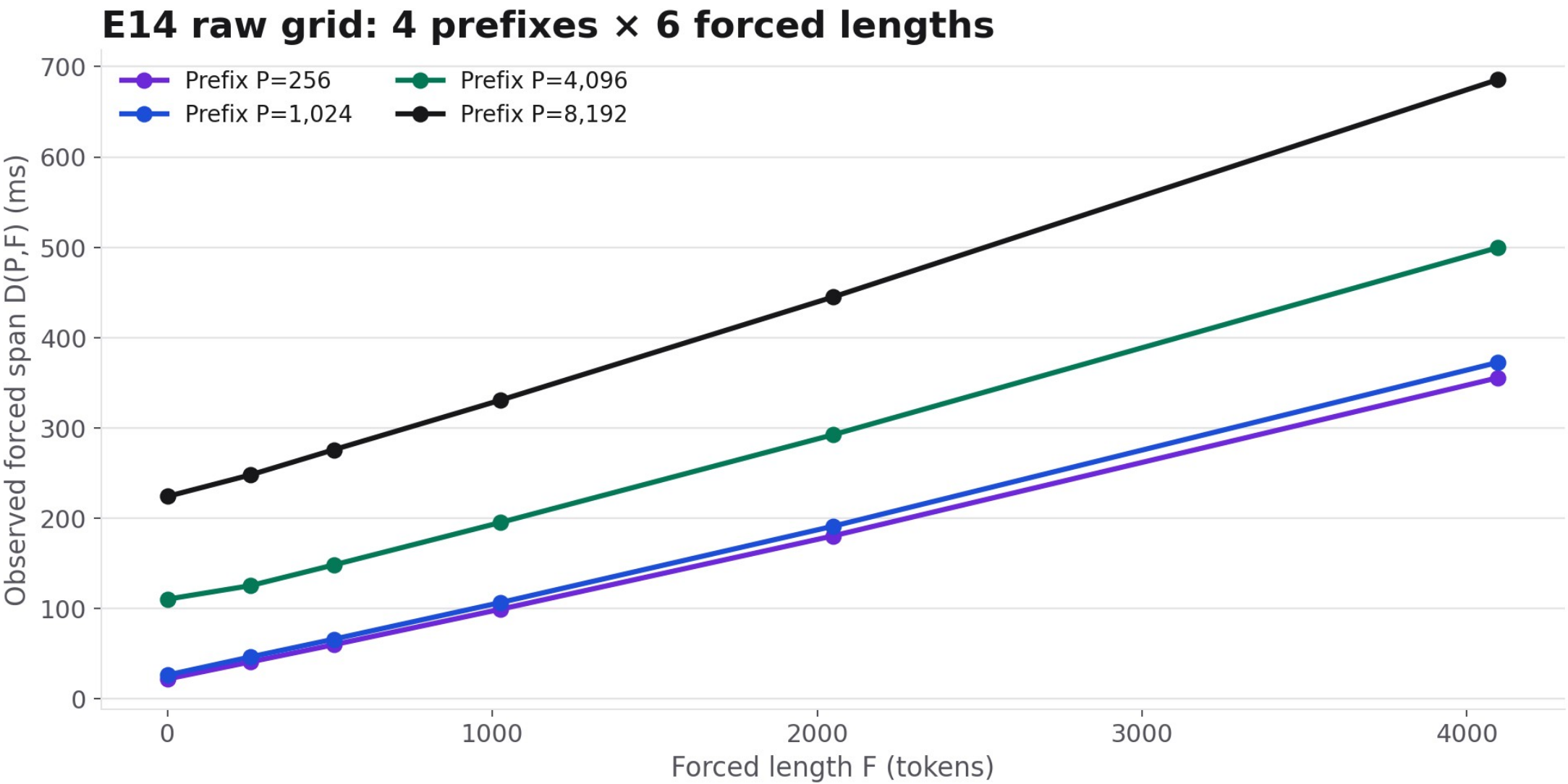
RESUME LANES

- 01** A — force-decode Lhist + Lnew in decode; no reload
- 02** B — reload Lhist KV; force-decode only Lnew
- 03** C — prefill Lnew over resident Lhist; transfer the completed KV state to decode

The experiments do not all exercise the same kernel path

EVIDENCE	KERNEL / PATH	SWEEP	QUALIFICATION
E5	pdSeparate decode KV ingress adaptor/injector	P=256...8192	Direct device TSC
E9	2×4 force-decode pipeline; 28 layers	P=256; F=1...20224	Direct device TSC
E14	Same force-decode model config	4 prefixes × 6 F	n=1; 24/24 pass
E10/D	Reload path + post-ingress force decode	H=512...8192	Two direct totals
E12a	S6a resident-prefix prefill, e0a19fc	Two 8192-context cases	Screening; not exact pdSeparate
E6	Prefill transpose + switch-gather egress	1...32 chunks	Not inverse-equivalent to E5

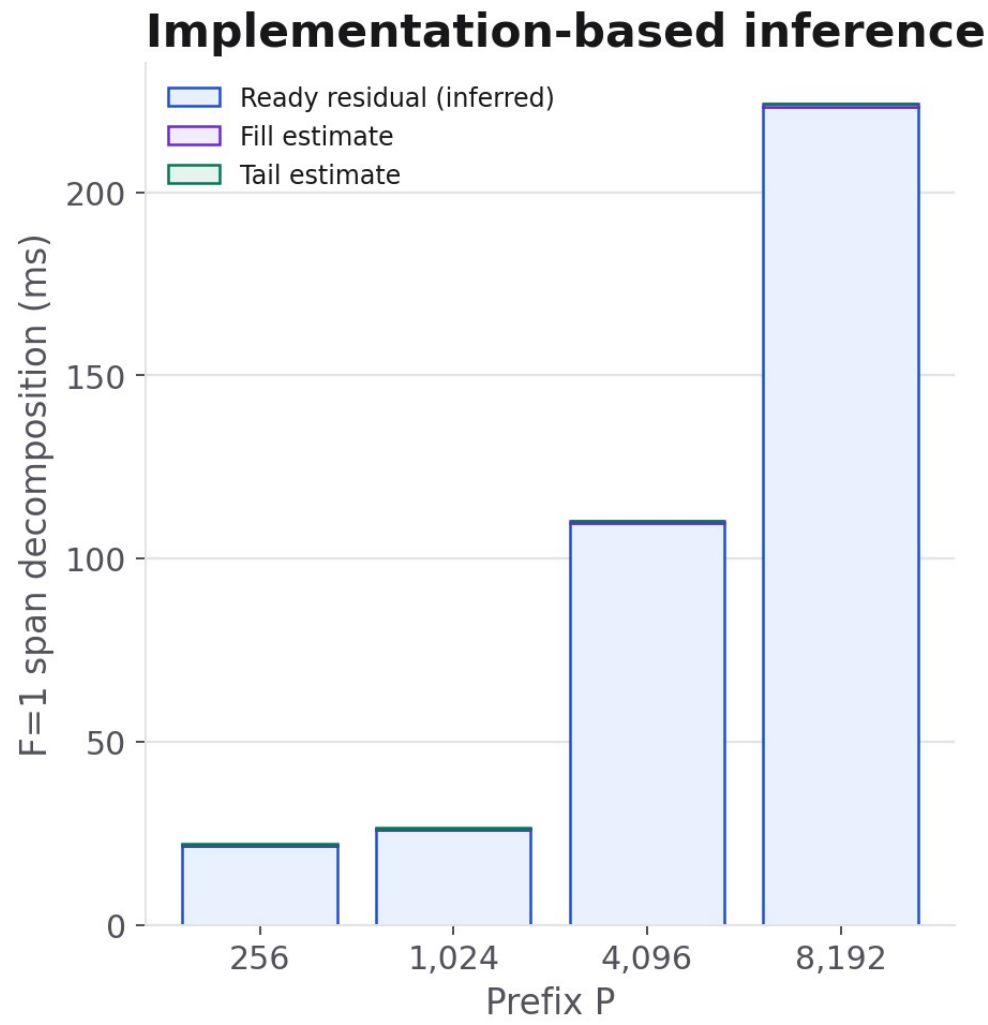
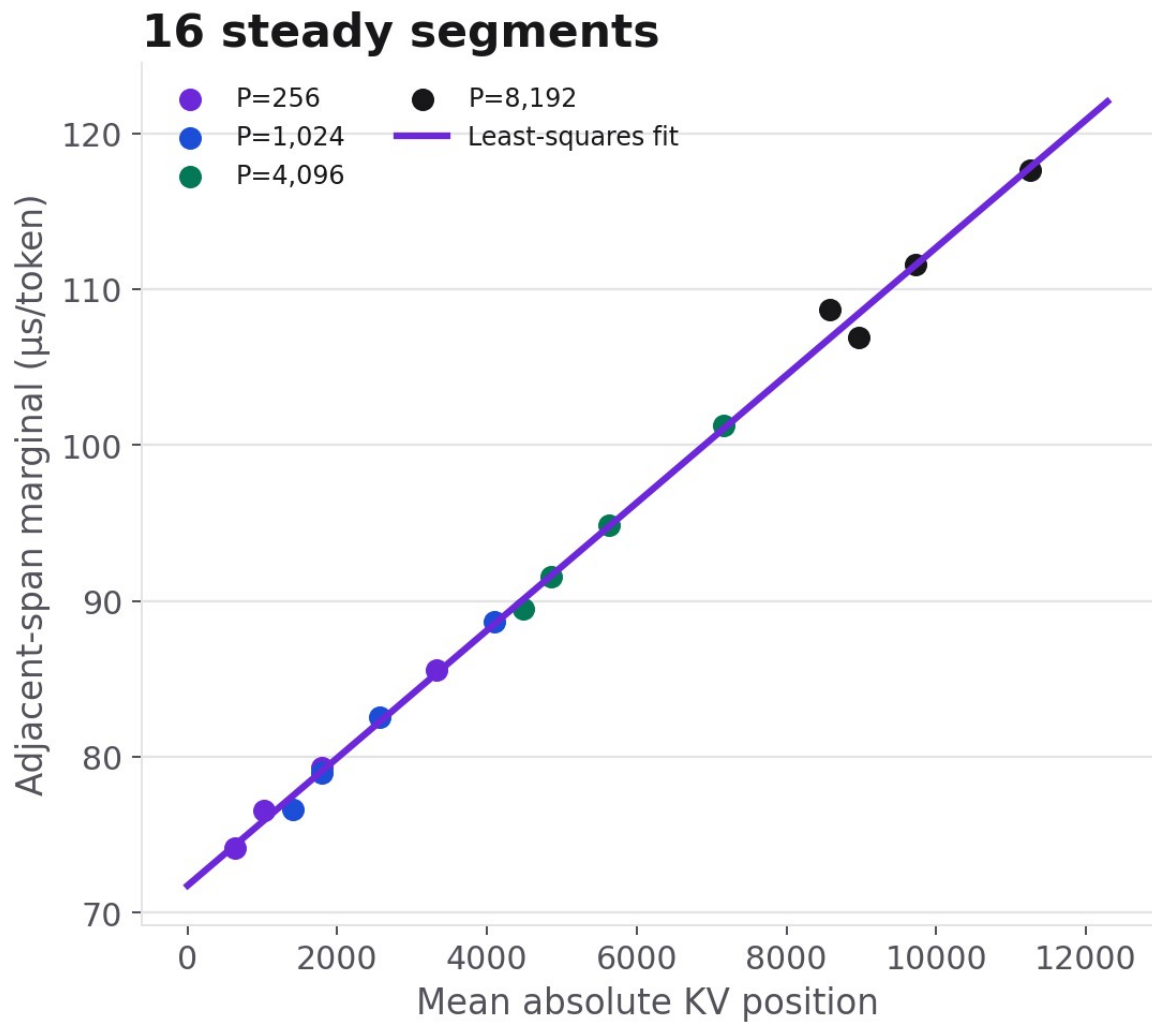
All 24 force-decode spans across prefix × forced length



RAW — $P=\{256,1024,4096,8192\}$, $F=\{1,256,512,1024,2048,4096\}$; real Qwen3-1.7B on WSE-3, $n=1$.

Startup anchor + position-dependent intervals

Derived views — not additional direct measurements



$D(P,F)=D(P,256)+\sum \text{II}(q)$, where 16 steady segments give $\text{II}(q)=71.745 \mu\text{s}+4.093 \text{ ns}\cdot q$. Ready/fill/tail are inferred.

Complete force-decode equations

$$D(P, F) = D(P, 256) + \sum_{q=P+256}^{P+F-1} ll(q), \quad F \geq 256$$
$$D(P, F) = D(P, 256) + (F - 256)a + \frac{b(F - 256)(2P + F + 255)}{2}$$

Equivalent closed form for the same discrete sum

STEADY INTERVAL

$$ll(q) = a + bq$$

$$a = 71.745198 \mu s$$

$$b = 0.004093307 \mu s/position$$

PREFIX-SPECIFIC STARTUP ANCHOR

$$D(P, 256) = d_k + \frac{P - p_k}{p_{k+1} - p_k} (d_{k+1} - d_k)$$

$$P \in [p_k, p_{k+1}]$$

$$p_k \in \{256, 1024, 4096, 8192\}$$

MODEL SCOPE

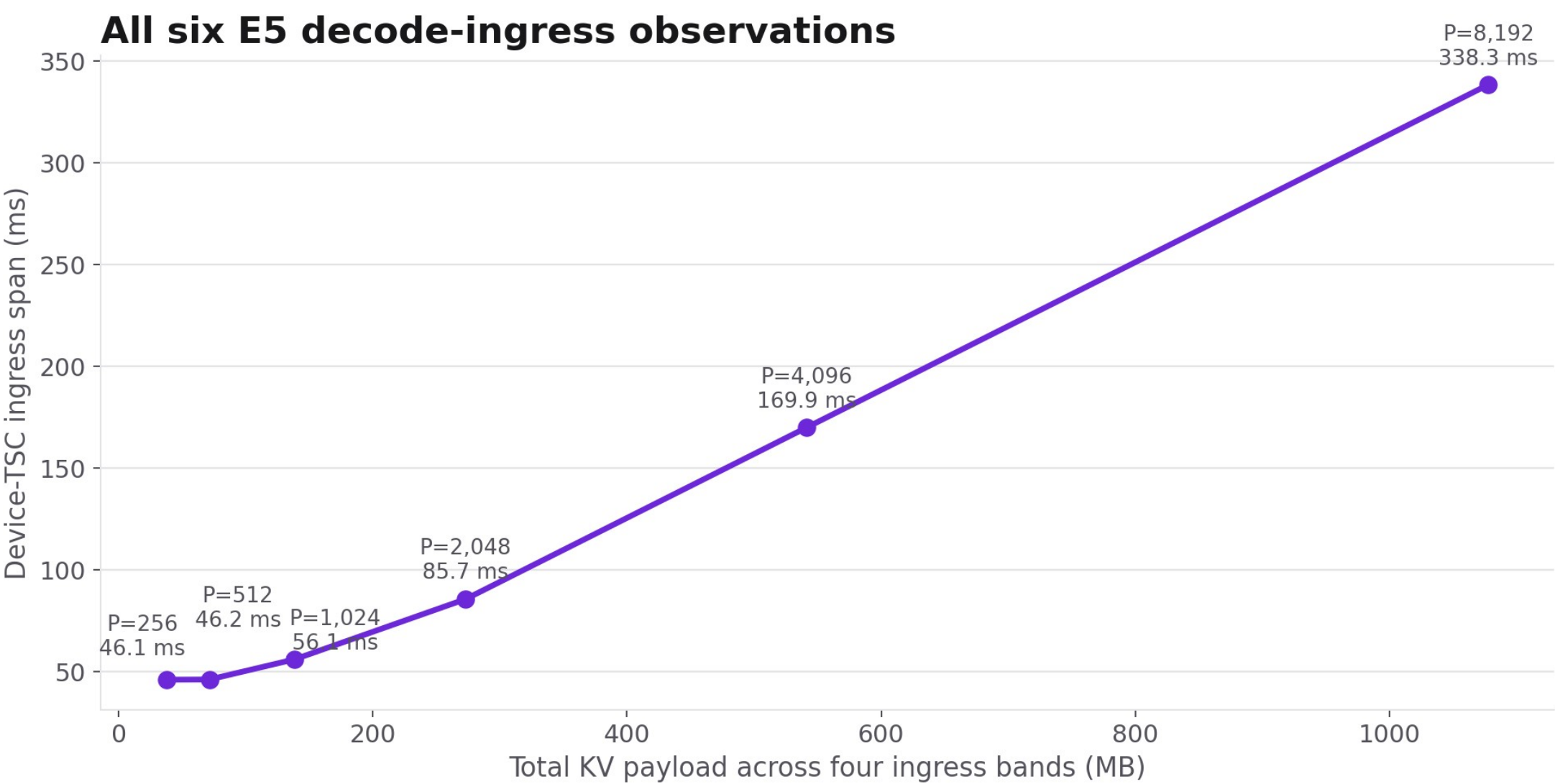
Anchor includes readiness

F=1 is not a universal
offset to subtract

E14: n=1

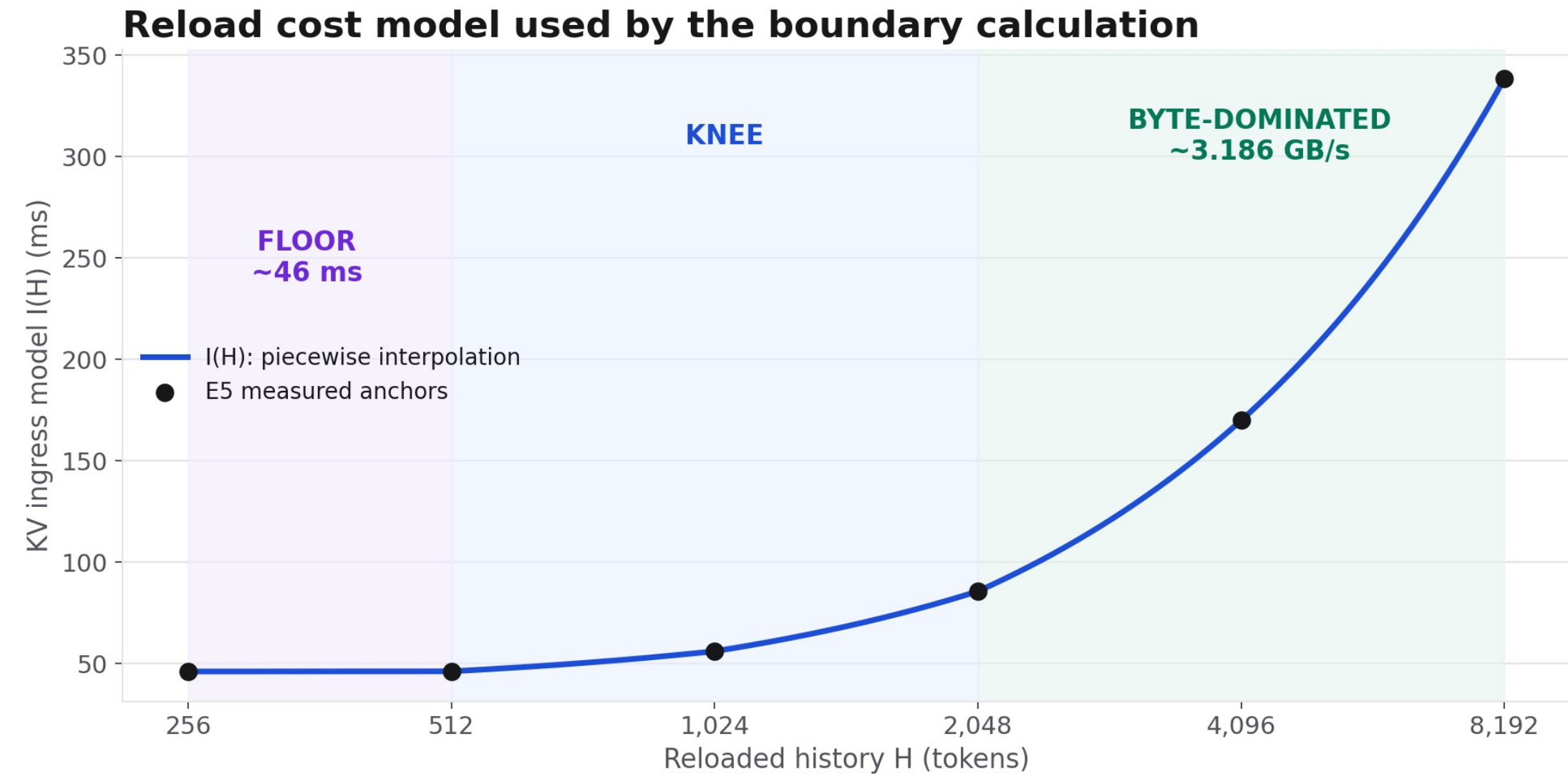
The discrete sum and closed form are equivalent. D(P,256) is interpolated between measured prefix anchors; F=1 is not subtracted as a universal offset.

Decode KV ingress: six raw measurements



RAW — six device-TSC observations on the pdSeparate ingress path, spanning the floor, knee, and byte-dominated region.

Reload = floor + payload term



$I(H)$ uses piecewise-linear interpolation of the six E5 anchors; the saturated region approaches 3.186 GB/s aggregate.

Lane B reload equations

$$B_{\Delta}(S, H, L_{new}) = I(H) + D(S + H, L_{new})$$

$$A(S, H, L_{new}) = D(S, H + L_{new})$$

Lane A reference used at the crossing

$$B_{full}(S, H, L_{new}) = I(S + H) + D(S + H, L_{new})$$

INGRESS MODEL

$$I(h) = t_k + \frac{t_{k+1} - t_k}{h_{k+1} - h_k}(h - h_k)$$

$$h \in [h_k, h_{k+1}]$$

$$h_k \in \{256, 512, 1024, 2048, 4096, 8192\}$$

FORCED DELTA

$$D(S + H, L_{new})$$

Reuse Lane A's startup
+ steady-II model
at absolute prefix S+H

VARIABLES

S: starting prefix

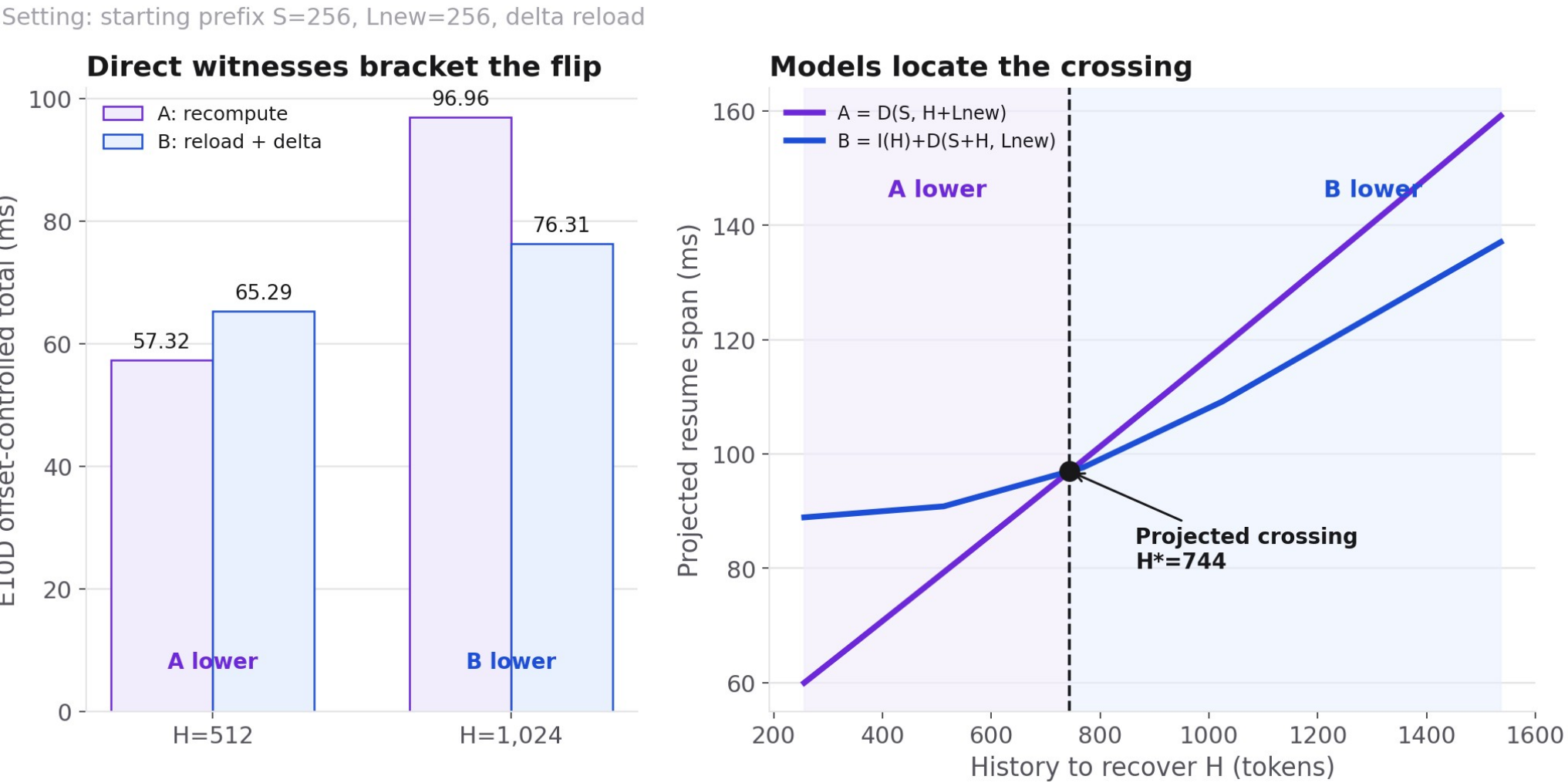
H: history to recover

L_{new}: new context

I: measured KV ingress

Delta reload pays I(H); full-context reload pays I(S+H). Both then force-decode L_{new} using the Lane A model at absolute prefix S+H.

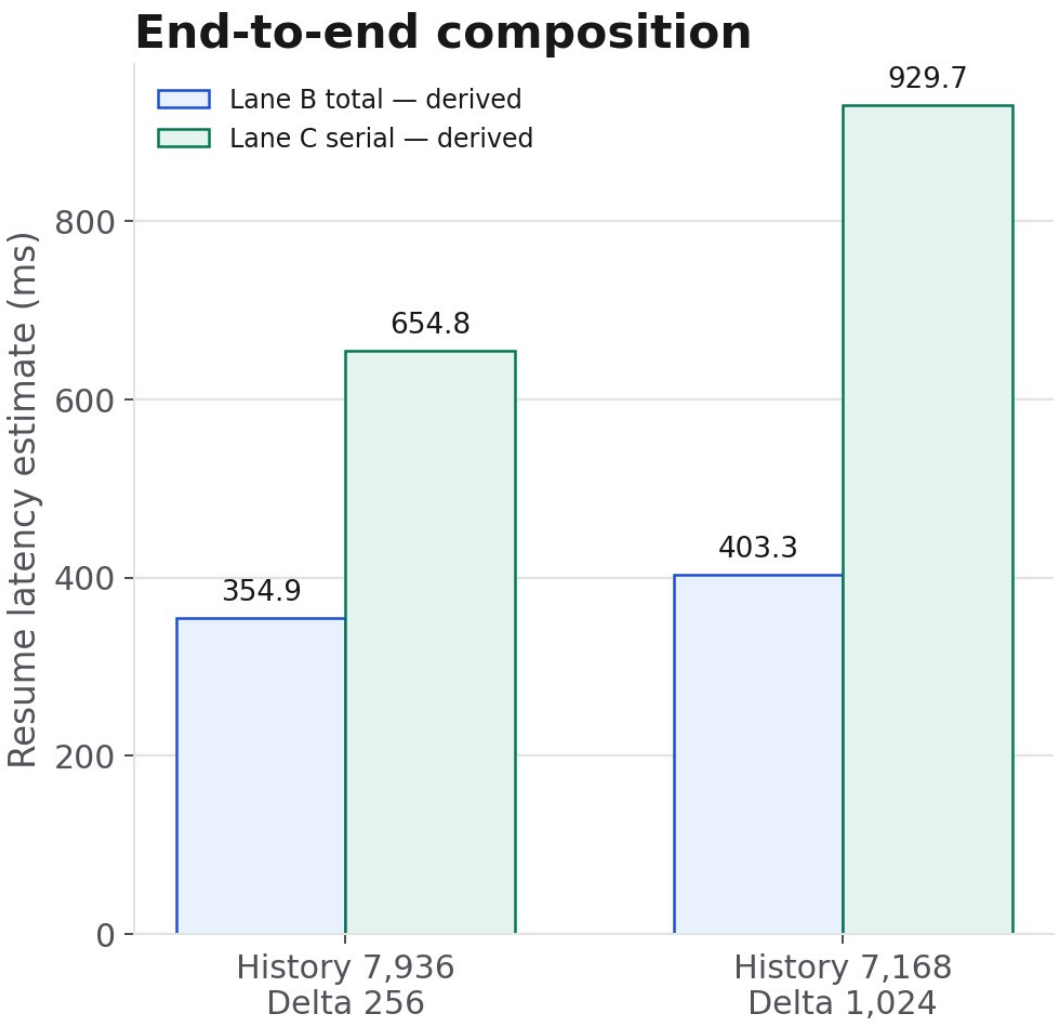
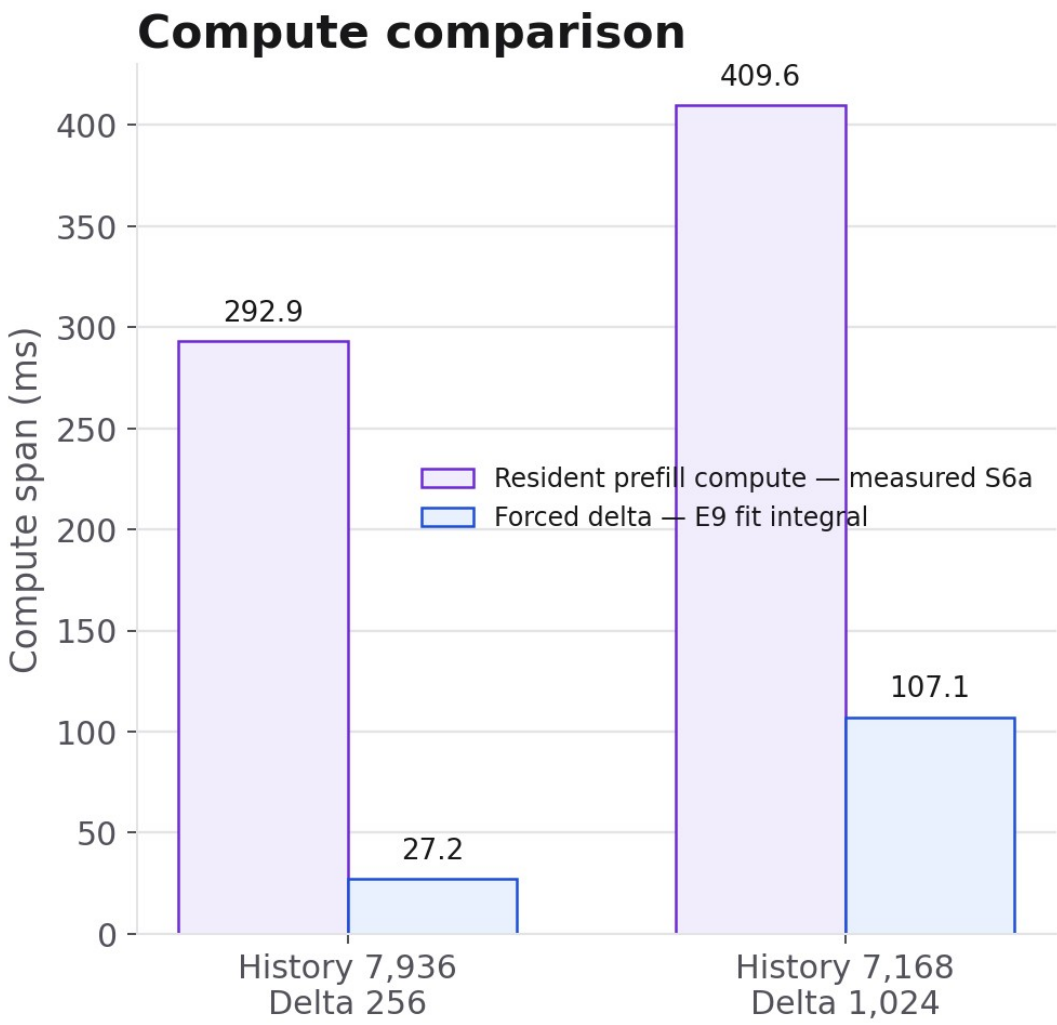
The measured flip and the projected crossing



Left: E10D offset-controlled totals. Right: E14+E5 observed-span projection gives $H^*=744$ (S=256, Lnew=256, delta reload); compare winner/crossing, not absolute y-values.

Two resident-prefix prefill cases at total context 8,192

E12a is a screening run on S6a e0a19fc, not exact pdSeparate



MEASURED + DERIVED — direct S6a prefill compute at total context 8,192; forced and lane totals combine E9/E5/E6 evidence.

WHAT THE EVIDENCE SUPPORTS

A conditional policy, with explicit gaps

STATEMENT	EVIDENCE STRENGTH	SCOPE / CAVEAT
Ingress has a ~46 ms floor	Direct measured	pdSeparate decode ingress
Forced steady cost grows with position	Measured + fit	16 E14 steady segments
A/B winner flips at 512 → 1024	Direct measured	Lnew=256 resume setting
Crossing moves with S and reload policy	Model-derived	Direct witnesses pending
Current Lane C loses	Screening	S6a path; exact port pending
Full eviction round trip favors B	Unknown	Decode egress E13 missing

Only run experiments that can change the decision

E14	Add a same-PE first-Z timestamp; separate readiness from first-token fill.
E14	Sweep $F=\{1,16,32,64,128,256\}$, alternate order, and obtain $n=2$.
E10	Directly witness $S=1024$ and $S=4096$ boundary neighborhoods.
E12a	Port to exact pdSeparate only if final confirmation is required.
E13	Decide whether full eviction round-trip latency is required for the policy claim.